

Smart Street Lighting System

IoT-Based Design and Implementation

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National Higher Engineering School of Tunis · 2024–2025

End-Year Project

The Problem

215 TWh

global yearly street lighting consumption

6%

of total world electricity

Always-ON

no motion awareness in traditional systems



Fixed schedules

Lamps run at full power all night, regardless of whether anyone is on the road.



Reactive maintenance

Faults are only found during physical inspections — sometimes days after they occur.



No intensity control

Traditional LDR systems switch lamps fully on or off — no intermediate dimming possible.

Our Solution

An IoT system that automatically adapts street lamp brightness to real-time conditions — monitored and controlled remotely via smartphone.



Auto Day/Night

LDR sensor detects ambient light — lamps switch automatically without any timer or manual input.



Motion-Triggered

PIR sensors detect vehicles and pedestrians — lamp jumps from 25% standby to 100% instantly.



Progressive Light

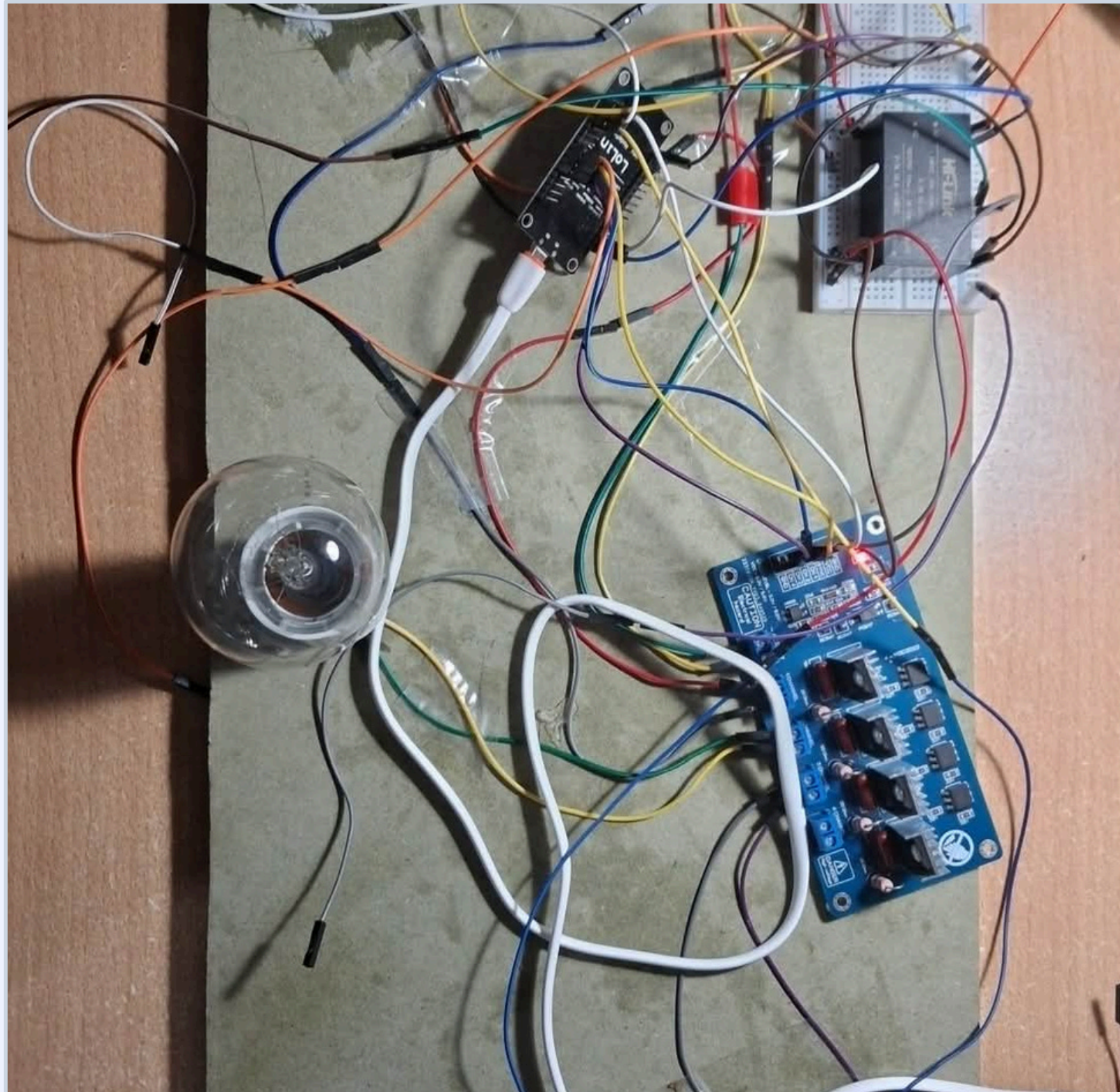
A wave of full brightness follows the vehicle — only the active zone is fully lit at any time.



Remote Dashboard

Node-RED dashboard on any smartphone — monitor all lamps, override manually, track energy.

Hardware Components



01

ESP8266 NodeMCU

Main microcontroller · Wi-Fi · MQTT client

02

HLK-PM01

220V → 5V DC isolated power supply

03

TCRT5000 PIR × 3

Motion detection — one per lamp zone

04

LDR + 10kΩ divider

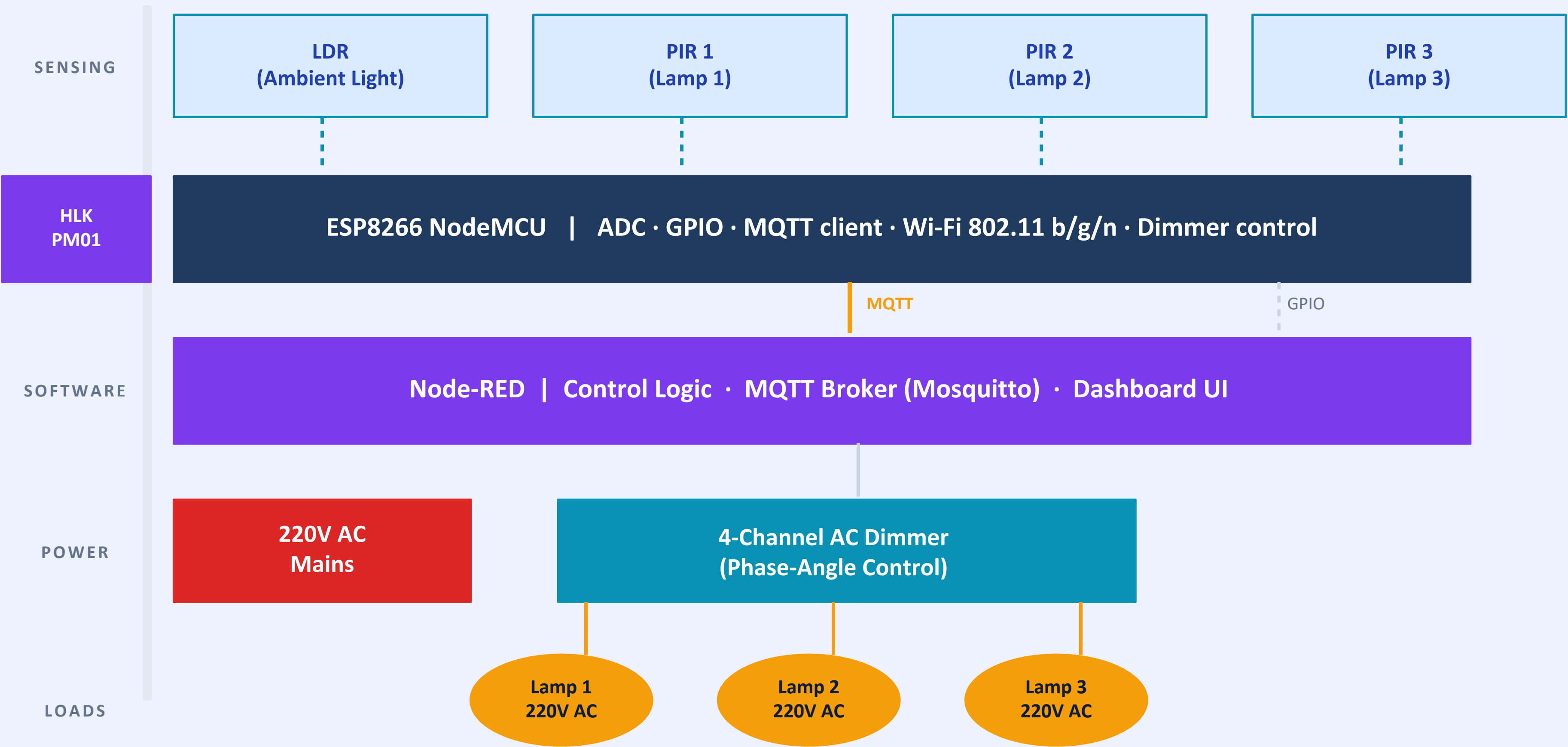
Ambient light sensing — day/night detection

05

4-Channel AC Dimmer

Phase-angle control · TRIAC per lamp channel

System Architecture



Software Stack

Arduino IDE

ESP8266 Firmware

Written in C++ with ESP8266 board support package

ESP8266WiFi.h — Wi-Fi connection management

PubSubClient — MQTT publish & subscribe

RBDdimmer — zero-cross interrupt + TRIAC timing

Narrow scope: sensors, dimmer, MQTT exchange

Node-RED

Control Logic + MQTT

Flow-based visual programming on Node.js

Mosquitto MQTT broker on port 1883

LDR threshold → night/day mode flag management

PIR sub-flows with 30s hold timer per lamp

Publishes brightness commands to ESP8266

Dashboard

5 Pages — Browser UI

Mode Automatique — sensor states + gauges

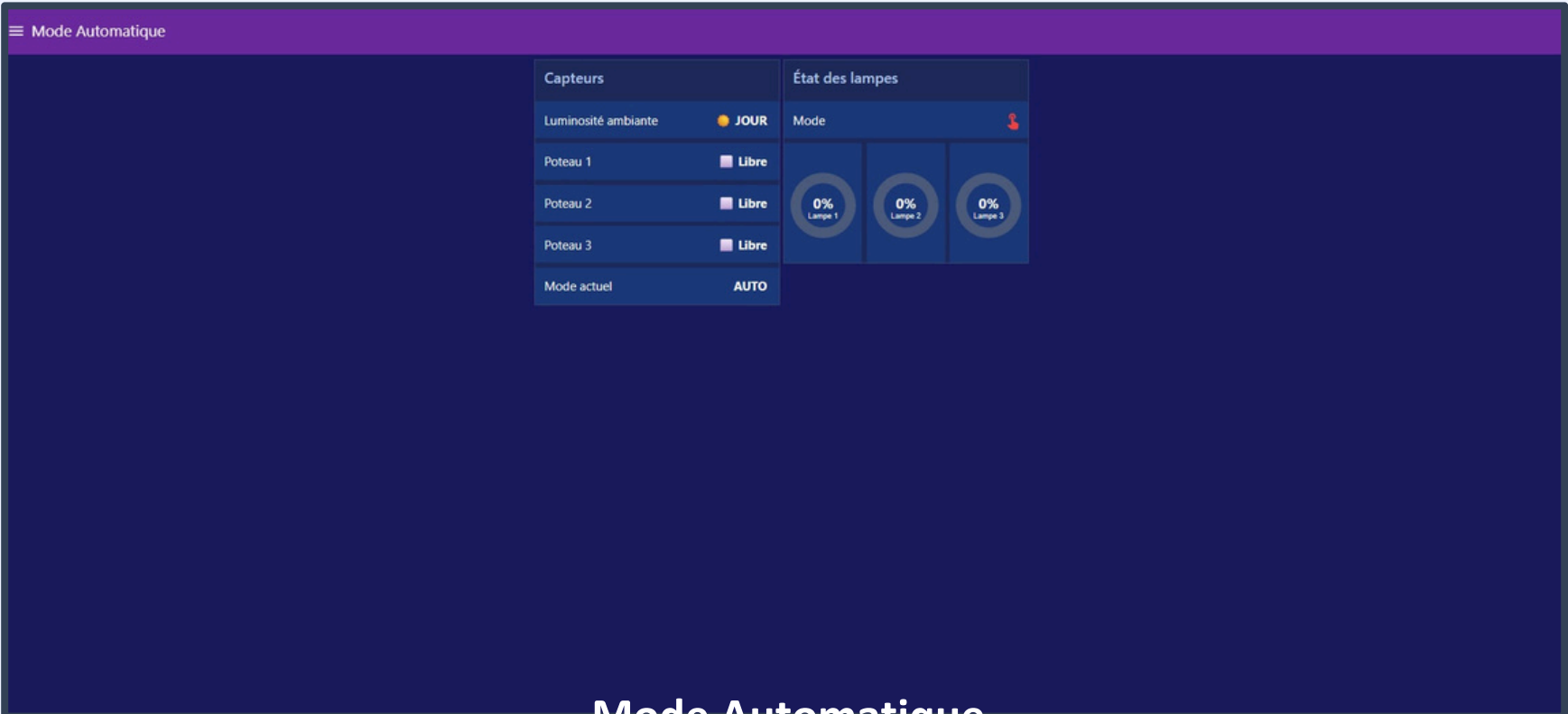
Mode Manuel — per-lamp ON/OFF buttons

Courbes — real-time brightness charts

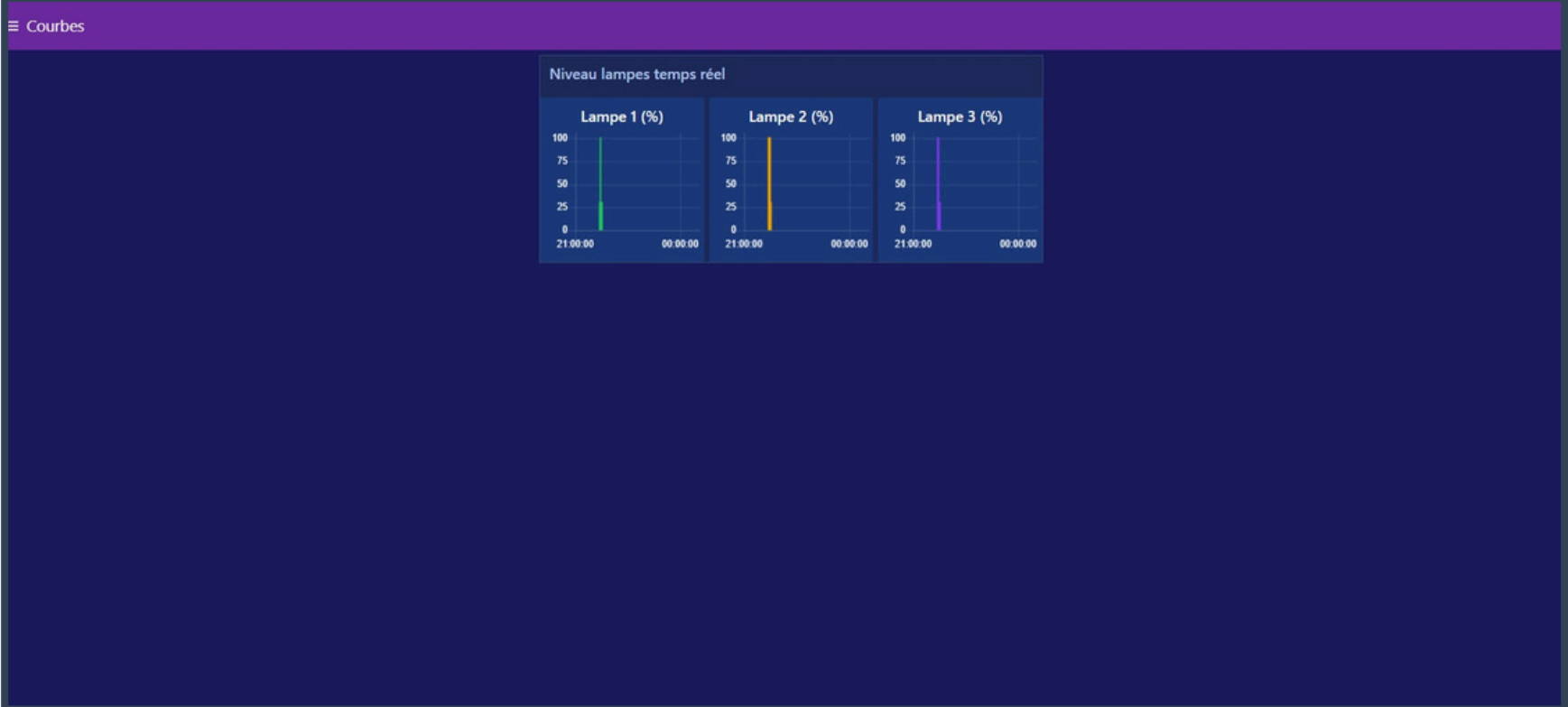
Activité PIR — detection timeline

Historique — passage counter

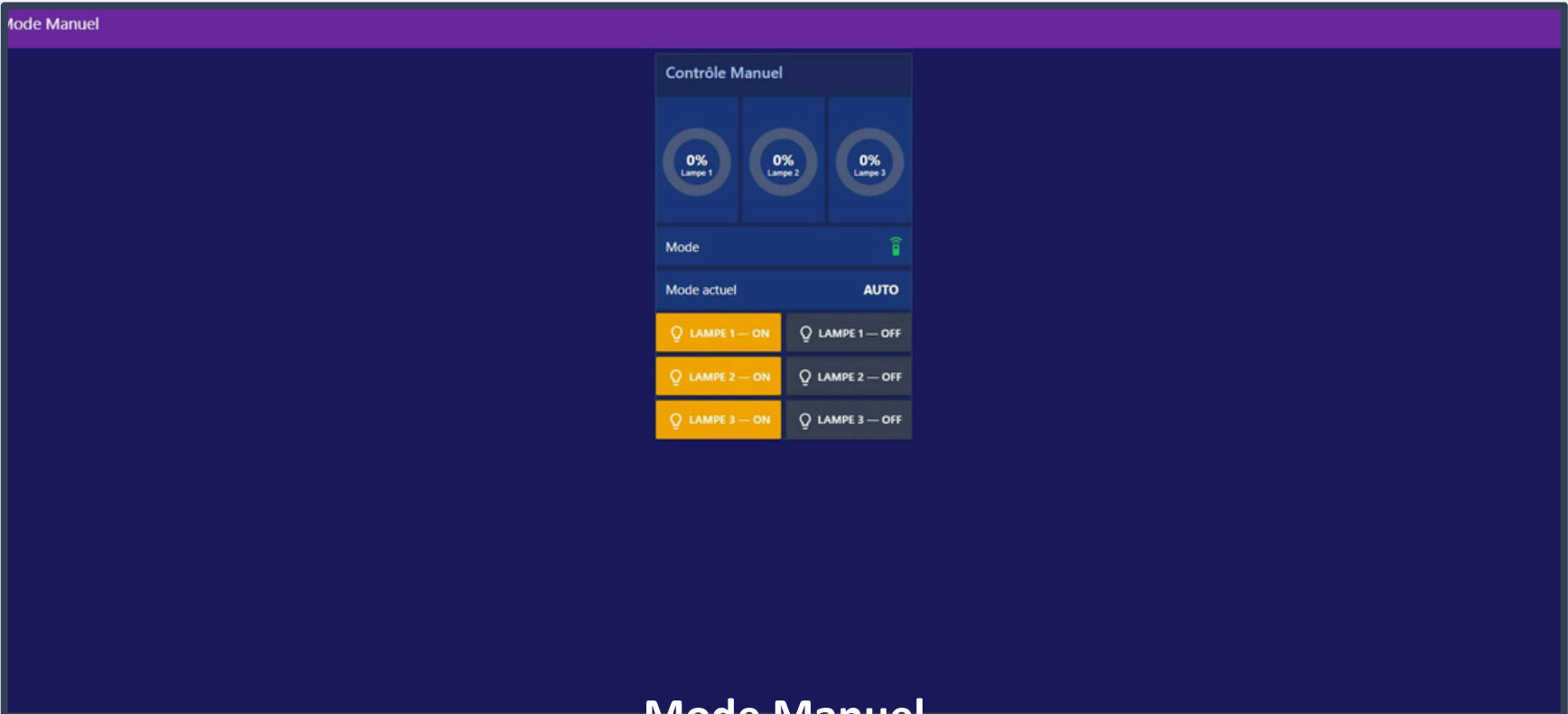
Live Dashboard — Node-RED



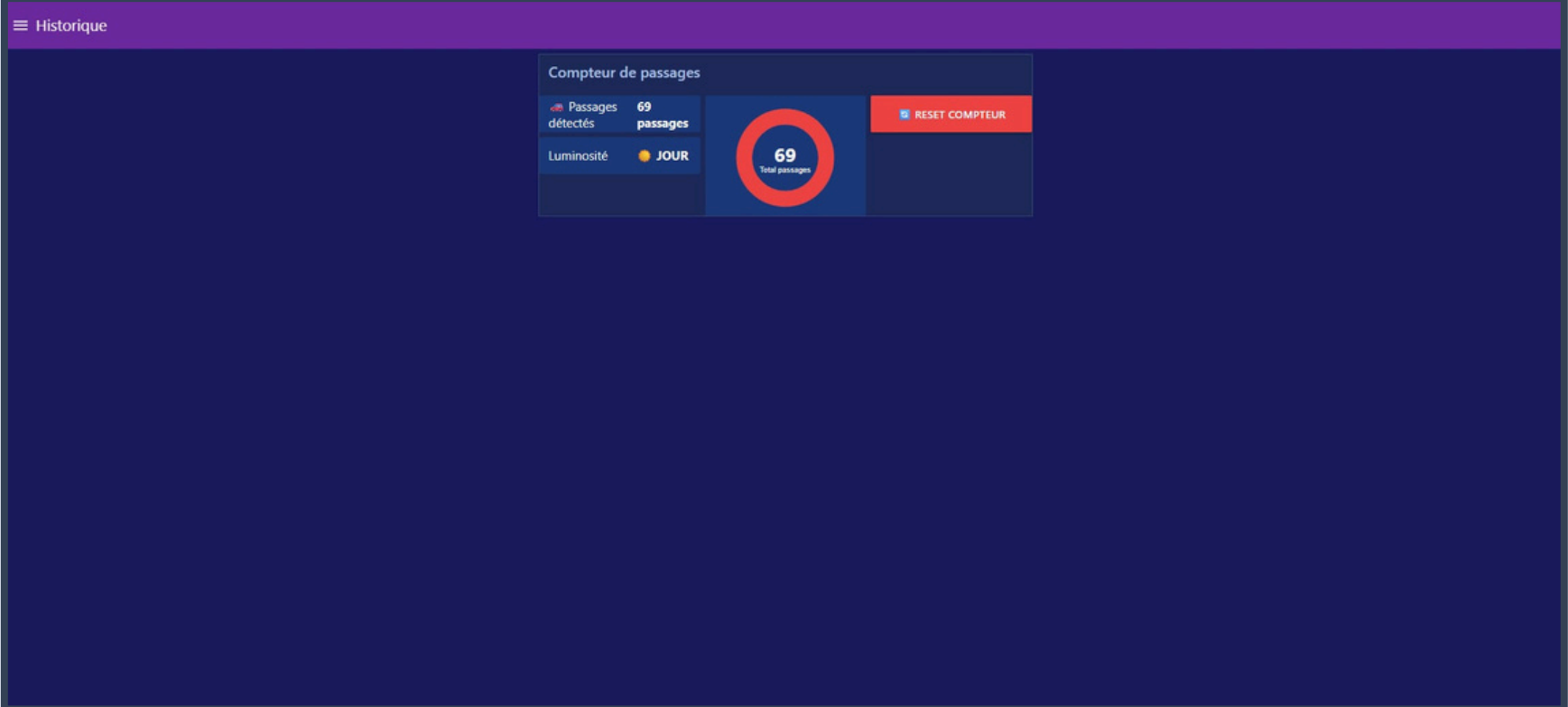
Mode Automatique



Courbes temps réel



Mode Manuel



Activité PIR & Historique

Validation Results

#	Test	Expected	Result
01	Day/Night detection via LDR	0% day · 25% standby at night	✓ fonctional · delay < 2 s
02	Motion detection (PIR)	Lamp rises to 100% on trigger	✓ fonctional · avg 200 ms
03	Progressive illumination	Only active zone at 100%	✓ Correct on walks
04	Manual dashboard override	Instant ON/OFF · AUTO resumes	✓ Actional < 1 s

1200 Wh

Always-ON
(baseline)

300 Wh

Standby only
25%

440 Wh

Mixed mode
25% + 20% active

75%

Max saving
(standby)

63%

Realistic saving
(mixed)

3 × 40 W lamps · 10-hour night · Values estimated from rated lamp power

Limitations & Future Work

⚠ Current Limitations

PC dependency

Node-RED host must stay online. If the PC goes offline, control stops and the ESP8266 defaults to its last state.

PIR in warm weather

Reduced thermal contrast on hot days causes occasional false positives. Sensitivity adjustment helps but doesn't eliminate the issue.

No persistent storage

Dashboard data is lost after a restart — no historical energy logs are kept between sessions.

→ Future Improvements

Raspberry Pi as host

Deploy Node-RED on a Raspberry Pi for 24/7 operation with no dependency on a laptop.

Radar motion sensors

Replace PIR sensors with microwave radar modules for reliable detection in any temperature condition.

Cloud data logging

Log energy data and passage counts to a cloud database for long-term historical analysis and reporting.

Thank You

A low-cost, open-source IoT system that saves up to 75% of street lighting energy — controlled from any smartphone.

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Questions?